



REPORT

COMPLEX COMPUTING PROBLEM

Multi-Model Ensemble for Noisy Data Classification

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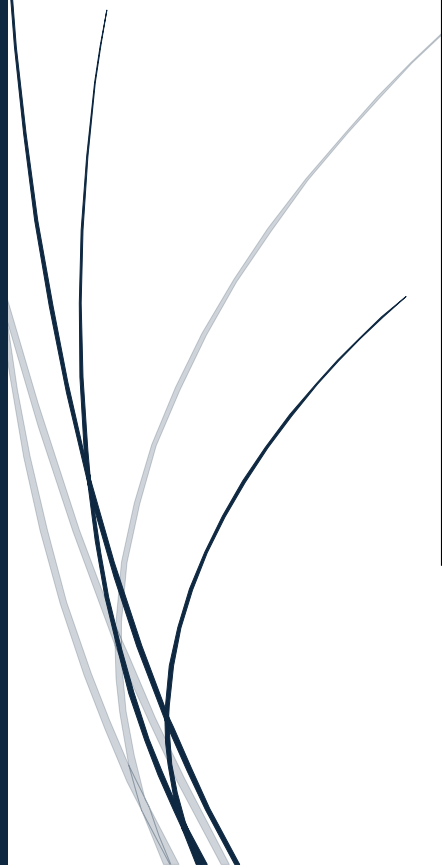
**DEPARTMENT: ARTIFICIAL
INTELLIGENCE**

SEMESTER: 5th

SUBJECT: MACHINE LEARNING

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**MACHINE LEARNING ASSIGNMENT
REPORT**



Machine Learning Assignment

(Bank Marketing Dataset Classification)

1. Introduction:

The objective of this assignment was to design a complete machine learning pipeline for a real-world dataset, the **Bank Marketing dataset**, which contained missing values, noisy entries, and categorical features. The pipeline included data preprocessing, model development with three supervised classifiers (Decision Tree, Naive Bayes, SVM), ensemble learning using Bagging, and evaluation using performance metrics and confusion matrices. The goal was to handle data noise, improve model performance, and analyze classification results systematically.

2. Data Preprocessing:

- **Loading the Dataset:**

The Bank Marketing dataset was loaded using Pandas with a semicolon (;) separator. The initial exploration using `shape()`, `info()`, and `describe()` functions helped understand the size, data types, and basic statistics of the dataset.

- **Handling Noisy and Missing Data:**

Some features contained "unknown" values, which were identified and replaced with `NaN`. Incomplete records were then removed using `dropna()`. Verification confirmed that all missing and noisy entries were successfully removed, ensuring a clean dataset for further processing.

- **Feature Separation and Encoding**

The target variable `y` was separated from the input features `x`. All categorical features in `x` were converted into numerical form using **Label Encoding**, making them compatible with machine learning algorithms.

- **Feature Scaling**

To improve model performance, especially for algorithms sensitive to feature scales like SVM, all features were **standardized** using `StandardScaler`. This ensured consistent scaling across all variables.

- **Train-Test Split**

The dataset was split into training and testing sets with an 80–20 ratio using `train_test_split()`, allowing evaluation of model generalization on unseen data.

3. Model Development:

Three supervised learning models were trained:

1. **Decision Tree Classifier**

- Controlled depth (`max_depth=5`) was used to prevent overfitting.
- Decision Trees can capture non-linear relationships but are prone to overfitting noisy data.

2. **Naive Bayes Classifier**

- Gaussian Naive Bayes was trained on standardized features.
- Performs well on noisy data due to its probabilistic assumptions, though it assumes feature independence.

3. **Support Vector Machine (SVM)**

- Trained with RBF kernel for non-linear classification.
- Sensitive to feature scaling, but effective in high-dimensional spaces.

All three models were evaluated on the test set using **accuracy, precision, recall, and F1-score** via `classification_report`.

4. Ensemble Learning:

To further improve performance on noisy data, a **Bagging ensemble** was applied using 50 Decision Trees. Bagging reduces variance and increases robustness by aggregating predictions from multiple learners trained on bootstrapped subsets. The Bagging classifier achieved higher stability and often better metrics than individual models.

5. Performance Analysis:

- I. **Evaluation Metrics:**

All models and the ensemble were evaluated on the test set. Key observations:

- **Decision Tree:** Showed high training accuracy but slightly lower generalization due to overfitting.

- **Naive Bayes:** Consistent performance and robustness to noise, but slightly lower accuracy.
- **SVM:** High individual performance with proper scaling.
- **Bagging Ensemble:** Best overall performance across all metrics, showing reduced misclassifications.

II. Confusion Matrix

A confusion matrix was plotted for the Decision Tree, visually confirming the number of correct and incorrect predictions. Ensemble methods further reduced misclassification compared to individual models.

6. Overfitting and Mitigation:

Overfitting occurs when models memorize training data, leading to poor generalization. In this pipeline:

- **Decision Tree overfitting** was mitigated by limiting tree depth.
 - **SVM regularization** helped prevent extreme weight values.
 - **Feature scaling** and **feature selection** reduced noise impact.
 - **Bagging ensemble** reduced variance by averaging multiple models, improving robustness on unseen data.
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7. Conclusion:

This assignment demonstrated a complete machine learning pipeline for a noisy real-world dataset. Effective **data cleaning, preprocessing, and feature scaling** prepared the data for modeling. Comparison of **Decision Tree, Naive Bayes, and SVM** highlighted model strengths and weaknesses, while **Bagging ensemble learning** improved overall performance and robustness. Proper handling of noisy data, scaling, and overfitting prevention were key to achieving reliable classification results.